

Credibility assessment of CFD models for intravascular catheter trackability and flow resistance under ASME V&V 40

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Abstract

Computational fluid dynamics (CFD) is frequently used to screen intravascular catheters before bench testing, but decision makers need transparent evidence that selected models are credible for their intended role. This article applies the ASME V&V 40 risk-informed credibility framework to two CFD models that predict surrogates of trackability and flow resistance for a generic family of over-the-wire peripheral catheters (4–6 Fr). Model S is a steady-state Navier–Stokes solver that computes annular pressure drop, near-tip shear, and lateral hydrodynamic side-loads in prescribed tortuous segments. Model T is a transient solver with moving boundaries that computes time-resolved axial drag and lateral side-load on an advancing catheter tip negotiating bends and stenosis surrogates. We organize the assessment by four mechanisms: axial annular pressure-drop governing resistance; Dean-type secondary flows at bends; jetting from tip and side-ports with local recirculation; and hemorheology-dependent viscosity effects. The context of use is pre-bench design screening to rank catheter variants by predicted lateral side-load, integrated push drag, and pressure drop (ΔP) for prioritization into benchtop testing. The model risk is medium: the simulations influence design gating but do not replace mandatory bench or clinical evidence. We score five credibility factors on a 0–5 Credibility Index (CI): parameter evidence; numerical discretization and iteration control; comparison to bench reference; decision linkage of reported metrics; and representativeness of test articles and phantoms. For annular resistance (M1), Model S achieved median absolute percentage error of 4.8% versus a 42-point bench dataset, with grid-convergence indices below 0.8% for ΔP ; Model T achieved 5.4% error for time-averaged ΔP with time-step uncertainty under 1.2%. For Dean-type side-loads (M2), Model T reproduced peak lateral forces within 8.9% (95% limits ± 1.9 mN) during standardized pushes; Model S under the same geometries had 16% error (± 3.1 mN) due to its steady-state assumption. For jetting and recirculation (M3), Model T matched asymmetry in side-loads within 10.7% across tip-only and side-port injection, while Model S exhibited a systematic underprediction (18%). For hemorheology (M4), both models reproduced the ΔP increase from a Carreau–Yasuda analog relative to Newtonian fluid within one percentage point; lateral-force changes with viscosity were captured to within 12%. The selected quantities of interest (QOIs) mapped to go/no-go gates: designs passing the 12 mN side-load and 150 Pa at 200 mL/min thresholds on the bench were the same designs passing the CFD criteria in 93% of 28 prototypes. The benchtop phantoms covered the interquartile tortuosity of 176 CTA-derived peripheral vessels and matched coated-shaft friction within 12%. Two model-specific scorecards summarize factor-wise CI per mechanism (20 rows each). We discuss implications for design decisions, limits of generality, and priorities for further validation. Within the stated context and risk, Model S is credible for resistance ranking and qualitative side-load trends; Model T is credible for ranking trackability surrogates and resistance with transient advancement.

Keywords: catheter CFD, V&V 40, trackability

1. Introduction

Hydrodynamic forces acting on intravascular catheters influence both the energy required to deliver therapy and the ease with which a device tracks over a guidewire through tortuous anatomy. Next to mechanical interactions such as friction and contact geometry, pressure-driven flows around and through the catheter shaft and tip generate axial and lateral loads that can bias steering, deflect the tip, and change the overall force balance perceived by the operator. In routine product development, engineers use computational fluid dynamics (CFD) to explore design variations early, long before bench prototypes are abundant. However, when CFD predictions inform design

triage across variants, a structured, risk-informed credibility assessment is needed to ensure that the models are adequate for their decision role.

The ASME V&V 40 standard provides a framework to align model credibility activities with the context of use and model risk for medical devices. The framework emphasizes that the sufficiency of evidence is proportional to the consequences of using the model to make a decision and the model's influence on that decision. This article applies that framework to two CFD models intended to screen a family of over-the-wire peripheral catheters (4–6 Fr) for trackability surrogates and flow resistance. We focus on four fluid-mechanics mechanisms that plausibly affect the target endpoints: axial annular pressure-drop,

secondary flows at bends, jetting at the tip and ports, and the influence of shear-thinning hemorheology.

We provide both detailed technical evidence and a compact, decision-relevant summary. First, we describe the device and decision need. Next, we define the context of use and model risk. We then introduce the two models, rationalize their model forms for the mechanisms of interest, and specify the quantities of interest (QOIs). We present parameter evidence, discretization and iteration controls, comparisons to benchtop reference data, and the representativeness of validation activities. We report one credibility scorecard per model, with five credibility factors scored across four mechanisms. We conclude with implications for design triage, limitations confined to the assessed factors, and priorities for further improvement.

General approach to V&V 40-aligned credibility — gradation.

- a. Narrative description of modeling goals without explicit linkage to decision-making or risk.
- b. Context of use stated; qualitative discussion of decision influence and consequences; ad hoc evidence collection.
- c. Explicit context of use; model influence and consequences assessed; evidence planned against selected credibility factors with traceability.
- d. Risk-informed mapping of evidence depth to decision impact; factor-specific criteria and thresholds; transparent reporting with mechanism-level granularity.

2. Regulatory background and motivation

Global regulatory bodies increasingly recognize in silico evidence as part of a total product life-cycle approach, provided that the modeling assumptions, verification activities, validation activities, and limitations are transparent and proportionate to the decision at hand. ASME V&V 40 formalizes this proportionality for medical devices by advocating a risk-informed approach rather than a fixed checklist. Our objective is to make the credibility arguments portable across internal gates and, if needed, suitable for external review, by clearly stating the mechanisms covered, the metrics reported, the relevance to design decisions, and the strength of the evidence.

In the development of peripheral intravascular catheters, two families of questions recur. First, which shaft and tip variants are likely to meet hydraulic resistance targets when used with standard pumps at nominal flow rates? Second, for tortuous paths and lesion surrogates, which variants are more likely to track without undue hydrodynamic bias on the guidewire? These are screening questions: the CFD results do not replace benchtop functional tests or clinical evaluation, but they do steer prototype selection, tooling priorities, and early design commitments. In this setting, the risk-informed approach prescribes establishing that the models capture the right physics at the right scale for the selected QOIs, are numerically trustworthy, and are shown to be sufficiently predictive against relevant laboratory comparators.

We adopt V&V 40 terminology, focusing on mechanisms, quantities of interest, and credibility factors relevant to this decision. We present a two-table summary of factor-wise confidence across mechanisms and models, reflecting both the heterogeneity of the physics and the different capabilities of a steady-state and a transient solver. Each table is positioned to be independently readable, matching common practice in regulatory submissions where model-specific evidence is presented in digestible pieces.

Model risk categorization under V&V 40 — gradation.

- a. No explicit model risk statement; implicit assumption of low impact.
- b. Model influence or consequence described qualitatively; no combined risk statement.
- c. Model influence and consequence both characterized; overall risk category assigned with rationale.
- d. Quantified model influence on decisions and bounded consequences; risk category mapped to required credibility activities with justification.

3. Device description and decision need

The device family under study is a set of over-the-wire peripheral catheters, nominal sizes 4–6 Fr, intended for use in lower-extremity and visceral arterial segments. The designs share a lubricious polymer-coated shaft, braided proximal reinforcement, and distal tapers with either a tip-only outlet or combined side-ports. Shaft outer diameters span approximately 1.33–2.00 mm (4–6 Fr), with lumen diameters 0.70–1.60 mm depending on function. Shaft stiffness gradients are mechanically important for kinking resistance and rail support, but in this work we isolate hydrodynamic effects by examining flows with and without shaft advancement.

The decision need is a pre-bench triage: among dozens of design variants, which ones are most likely to meet pressure-loss targets at set pump conditions and which ones minimize hydrodynamic lateral bias that may hinder guidewire tracking in tortuous vessels and near lesions? The development team uses these CFD outputs to sort designs into three bins prior to bench: proceed to prototype, redesign minor features, or hold. The bins are defined by thresholds on ΔP at 200 mL/min and on lateral side-load and integrated axial drag during a standardized push around bends and lesion surrogates. The CFD's role is influential but not determinative; benchtop functional testing remains the gate for design verification.

Decision linkage of modeling results — gradation.

- a. Outputs are interesting but not mapped to decisions or specifications.
- b. Qualitative mapping from outputs to design choices; no thresholds.

- c. Quantitative thresholds defined and tied to outputs; retrospective examples.
- d. Prospective, pre-specified thresholds with acceptance bands; decisions documented against model outputs.

4. Context of use (CoU) and model risk (ASME V&V 40)

We define the context of use as pre-bench design screening to rank catheter variants by predicted trackability surrogates (lateral side-load at bends, integrated axial drag during a standardized push) and hydraulic flow resistance (ΔP across a defined segment) for selection into benchtop testing. The simulations are not intended to predict clinical outcomes or to replace bench tests; instead, they prioritize designs for fabrication and testing. The model influence on the design gate is moderate because laboratory confirmation is run in parallel and overruling the model is routine when bench data disagree.

The consequence of an error in the simulations is limited to delayed iteration and suboptimal prototype selection; no patient harm can result directly because the modeled outputs do not influence labeling or clinical decision-making. Given the moderate influence and limited consequence, we designate the overall model risk as medium. Under V&V 40, medium risk supports a structured validation program against relevant comparators, sensitivity analysis of parameters that most affect the QOIs, and rigorous numerical verification to separate discretization uncertainty from model-form and parameter uncertainty.

Evidence depth mapped to medium risk — gradation.

- a. Minimal verification and validation; reliance on expert opinion.
- b. Basic solver checks and spot comparisons; limited documentation.
- c. Systematic numerical verification; planned validation against lab data; sensitivity analysis for key parameters.
- d. Comprehensive verification with error quantification; multiple validation datasets across operating space; uncertainty propagation into decision metrics.

5. Models under assessment

We evaluate two models. Model S is a steady-state, three-dimensional Navier–Stokes finite-volume CFD model with no-slip walls, either Newtonian or Carreau–Yasuda shear-thinning viscosity, and prescribed inflow and outflow boundary conditions. The computational domains represent arterial segments with curvature and torsion representative of lower-extremity femoropopliteal and iliac paths. The catheter is concentric or eccentrically offset by known clearances inside the vessel lumen, and either terminates with a tip outlet or includes

side-ports in the distal 10–20 mm. The model computes the annular pressure drop between the vessel wall and catheter shaft, shear stresses in the tip region, and lateral hydrodynamic side-loads on the catheter tip at steady inflow. The primary quantities of interest are ΔP across a defined test segment and the magnitude of lateral side-load at specified bends.

Model T is a transient CFD model with moving boundaries, in which the catheter is advanced over a fixed guidewire through a tortuous path and stenosis surrogates using a prescribed axial speed profile. A dynamic mesh method with local smoothing and remeshing preserves cell quality during advancement. The model computes time-resolved axial drag and lateral side-load on the distal tip, which vary as the tip negotiates curvature and enters/leaves constrictions. Quantities of interest are the cycle-averaged ΔP , the peak lateral side-load during each bend passage, and the integrated hydrodynamic drag during a standardized push equivalent to 40 mm of advancement at nominal 10 mm/s with dwell pauses at bends.

Both models use laminar flow assumptions. For the range of Reynolds numbers encountered in peripheral arteries with partial obstruction by a 4–6 Fr shaft and flows between 50 and 300 mL/min, transitional turbulence is unlikely; local Reynolds numbers in the annular gap and at the tip rarely exceed 900 for the geometries considered. For jetting from side-ports, we considered the potential for unsteady shear-layer oscillations; these were resolved by the transient model's time step and mesh resolution, and no turbulence modeling was employed.

Model form selection relative to mechanisms — gradation.

- a. Model form not justified against physics of interest.
- b. Model form qualitatively matches mechanisms; alternatives not considered.
- c. Model form justified against operating Reynolds and scales; alternatives discussed; limitations noted.
- d. Model form selected via analysis of dimensionless groups; alternatives tested for materiality; model form uncertainty bounded against comparators.

6. Mechanisms of interest and quantities of interest

We focus on four mechanisms that plausibly govern the target endpoints. M1 is axial annular pressure-drop between vessel wall and catheter shaft over a defined segment. M1 determines delivered flow at set pump conditions and contributes to the insertion force budget. The QOI is ΔP across a 120 mm test segment under nominal flows of 100–300 mL/min, for each catheter size and clearance condition. M2 is the emergence of Dean-type secondary flows in bends, generating lateral hydrodynamic side-load on the tip. M2 matters because side-loads can bias the tip relative to the wire when tracking through tortuosity. The QOI is the peak lateral side-load on the tip while traversing a 90-degree planar bend of radius 40 mm or a compounded S-bend with 35 and 45 mm radii.

M3 is jetting from the tip and side-ports with local recirculation. Ejecta from distal openings in proximity to a wall can create asymmetric pressures and vortices that perturb alignment near lesions. We define two outlet configurations: tip-only and a combined configuration with two 0.4 mm side-ports located 6 mm proximal to the tip at 30-degree offsets; nominal flow splits are 70/30 (tip/ports) at steady conditions in the absence of confinement. The QOIs are the asymmetric component of lateral side-load and the local shear stress on the tip region when adjacent to a stenosis surrogate (50% area reduction).

M4 is hemorheology-dependent viscosity. Blood's shear-thinning behavior alters both resistance and shear-driven lateral forces. We model this with a Carreau–Yasuda law fitted to a 45% hematocrit analog fluid: μ_{∞} 0.0035 Pa·s, μ_0 0.16 Pa·s, λ 8.2 s, n 0.64, a 2. We also use a Newtonian reference fluid of 3.5 mPa·s to mirror glycerol–saline analogs. The QOIs are the percent increase in ΔP and any change in peak lateral side-load between the Newtonian and non-Newtonian fluids under identical volumetric flows.

Definition and traceability of quantities of interest — gradation.

- a. Outputs not identified as QOIs; metrics ill-defined.
- b. QOIs named; measurement or computation described qualitatively.
- c. QOIs defined with units, locations, and time windows; traceable to decision thresholds.
- d. QOIs defined a priori with computational and experimental protocols; uncertainty on QOIs characterized and linked to acceptance criteria.

7. Credibility plan and scoring rubric (CI 0–5) mapped to ASME V&V 40 factors

We adopt a 0–5 Credibility Index (CI) scale, where 0 denotes no evidence and 5 denotes compelling, quantified evidence. We score five factors for each (model $\times$ mechanism) pair: parameter evidence (sources, bounds, and sensitivity of parameters that drive the QOIs), discretization and iteration controls (quantified numerical uncertainty), comparison to bench reference (predictive accuracy against relevant laboratory data), decision linkage of the reported metrics (strength of mapping from QOIs to go/no-go criteria), and representativeness of test articles and phantoms (how well the validation activities mirror the screening context). This yields 40 scored rows across two model-specific tables. We treat topics such as test conditions and operator-related variability separately as contextual notes without grading them, because they are handled during bench execution rather than being intrinsic to the CFD formulations.

Across the plan, we aim for mechanism-level granularity. For instance, numerical uncertainty in ΔP for annular flow is substantially smaller than that for peak side-load near recirculation; the scorecards reflect these differences. The scoring thresholds for CI=5 require both consistency of performance

across operating points and quantified uncertainties; CI=3 corresponds to moderate evidence with some gaps; CI=1–2 denotes weak or partial evidence.

The evidence plan was executed prospectively. Bench comparators were assembled with paired CFD predictions for design variants, sizes, and flow rates. Numerical verification studies were conducted before comparisons to avoid tuning mesh sizes based on hindsight. Sensitivity analyses were used to prioritize parameter measurement fidelity and to quantify the impact of realistic variability on the QOIs.

Credibility Index ladder (0–5) — gradation.

- a.
 - 0: No evidence or undocumented assumptions.
- b.
 - 2: Partial evidence; qualitative support; missing quantification or traceability.
- c.
 - 3: Moderate evidence; quantification exists but is incomplete or not general across operating space.
- d.
 - 5: Compelling, quantified evidence across relevant space; uncertainties bounded; traceability established end-to-end.

8. Model inputs — evidence and bounds for each mechanism (Model S and Model T)

The parameterization of geometry, materials, and boundary conditions was harmonized between the two models where feasible, with mechanism-specific elaborations. Vessel centerlines and lumen diameters were derived from a library of 176 CTA scans of peripheral arteries, segmented with semi-automated tools and manually quality-checked. The phantom paths used for both CFD and bench were designed to cover the interquartile range of tortuosity and curvature in the CTA library. For the primary 90-degree planar bend, radius was set to 40 mm (IQR 35–48 mm in the library); an S-bend used radii of 35 and 45 mm with 120 mm straight between bends. Vessel inner diameters in phantoms were 4.5, 5.5, and 6.5 mm, corresponding to quartiles of the target population. Dimensional tolerances for printed phantoms, verified by micro-CT, were ± 0.05 mm on lumen diameter and ± 0.3 mm on bend radius.

Catheter outer diameters were 1.33 mm (4 Fr), 1.67 mm (5 Fr), and 2.00 mm (6 Fr), verified with pin gauges (± 0.02 mm). Distal tapers were 10 mm long with a 3-degree angle. The guidewire was modeled as a 0.89 mm (0.035 in) cylinder. Shaft-to-wall clearances varied from 1.25 to 2.6 mm depending on size and vessel. The relative eccentricity in the annulus, induced by gravity in bench setups, was modeled as an offset of 0.2–0.5 mm in sensitivity runs; its first-order effect on ΔP under steady flow was below 2% but influenced lateral forces at bends by up to 8% depending on curvature.

Fluids included a Newtonian analog of 3.5 mPa·s at 37°C and a Carreau–Yasuda analog fitted to whole-blood rheometry at 45% hematocrit (as noted above). Dynamic viscosity of the Carreau analog at 100 s⁻¹ was 3.8 mPa·s, increasing to 11–14 mPa·s at

10 s1. Density was 1050 kg/m³ for both. Inlet flow rates were 50, 100, 200, and 300 mL/min, delivered by a gear pump with $\pm 1\%$ accuracy verified by a coriolis flowmeter in the bench, and mirrored as volumetric flow-rate boundary conditions in CFD. For the transient model, the standard push was 40 mm at 10 mm/s with 0.5 s dwells before and after each bend; accelerations and decelerations were limited to 100 mm/s² to avoid inertial overshoot in the drive system.

Sensitivity analyses used paired local derivatives and variance-based approaches. For ΔP in annular flow (M1) at 200 mL/min in a 5.5 mm vessel with a 6 Fr shaft, one-at-a-time perturbations of vessel diameter by ± 0.05 mm changed ΔP by 6.4%/+7.1%; viscosity perturbations by ± 0.2 mPa·s at 100 s1 changed ΔP by $\pm 3.6\%$. A Sobol' analysis across vessel diameter (± 0.05 mm), catheter diameter (± 0.02 mm), and viscosity (± 0.2 mPa·s) yielded first-order sensitivity indices of 0.62, 0.28, and 0.08, respectively, with the remainder due to interactions. For side-load in M2, bend radius and eccentricity dominated: ± 5 mm on radius shifted peak lateral force by 11%/+9%; eccentricity of 0.5 mm increased peak force by 7–10%. For M3, the tip/side-port flow split (nominally 70/30) varied by ± 8 percentage points across sizes; peak lateral force asymmetry was most sensitive to that split (first-order index 0.41), followed by tip-wall gap during stenosis passage (index 0.33). For M4, the Carreau parameters λ and n were varied within rheometer fit confidence intervals ($\pm 12\%$ and ± 0.03 , respectively); ΔP increase from non-Newtonian rheology varied by ± 1.6 percentage points across those ranges.

Parameter evidence and sensitivity — gradation.

- a.
Key parameters assumed; no sources or bounds.
- b.
Parameters sourced; tolerances known; limited sensitivity analysis.
- c.
Parameters measured or derived from population data; targeted sensitivity analysis identifies dominant contributors to QOIs.
- d.
Parameters with traceable metrology; uncertainty quantified; global sensitivity supports prioritization and uncertainty propagation.

9. Numerical discretization and iteration controls for each mechanism (Model S and Model T)

We conducted mesh and time-step refinement studies specific to each mechanism. Model S used unstructured tetrahedral meshes with prismatic boundary layers near the catheter and vessel walls. Near-wall y^+ was below 1 at all walls. For M1 and straight segments, three systematically refined meshes were used: coarse 1.2 million cells, medium 3.6 million, and fine 9.6 million. ΔP differences between medium and fine meshes were 0.5–0.8% across flows; extrapolated grid-convergence indices (GCI, 95% confidence) were 0.6% or less. For M2 and M3, tip and bend regions were locally refined to minimum edge lengths of 80

μm around the tip and side-ports; whole-domain counts were 2.4, 6.8, and 12.5 million cells for coarse, medium, and fine. Lateral side-load differences between medium and fine were 2.1–3.4% for steady inflow; GCIs were 2.7–3.8% depending on flow rate. Solver residuals were reduced to below 106 for mass and momentum, and steady-state monitors on ΔP and force components were converged within 0.1% over 500 iterations.

Model T used dynamic meshing with local remeshing triggered at cell skewness above 0.6 or aspect ratio above 20 near the moving tip. Static mesh counts (before advancement) were 1.8, 5.5, and 14 million cells for coarse, medium, and fine variants, with 70–120 μm resolution near the tip and ports. Time steps were 1.0, 0.5, and 0.25 ms. Time-step refinement showed that peak lateral side-load changed by 3.6% (coarse to medium) and 1.2% (medium to fine) in the 90-degree bend passage; integrated axial drag over 40 mm changed by 2.7% and 0.9%, respectively. For ΔP averaged over the push, time-step uncertainty was under 1.2%. At each time step, nonlinear residuals were reduced below 105, and temporal convergence was established by repeating sequences with 0.25 ms steps at select operating points. Conservation of mass was verified to within 0.05% per time step.

In both models, domain-length sensitivity was checked by extending inflow and outflow lengths by 2–4 diameters; ΔP changed by less than 0.3%, and lateral forces by less than 0.5 mN. For jetting cases in M3, outlet backflow stabilization was used with outflow sections extended 8 diameters to avoid reflections; sensitivity to the stabilization coefficient was negligible within the chosen range. Double-precision arithmetic was used throughout. No turbulence models were applied; a check with Smagorinsky–LES at the finest mesh and 0.25 ms time step for one high-flow, port-jet case in a 5.5 mm vessel showed less than 2% change in peak lateral force, supporting the laminar assumption for the studied conditions.

$$CI_{\{\text{score}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash \text{varepsilon}_{\{\text{rel}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\{\text{val}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

Discretization and iteration control — gradation.

- a.
Single mesh or time step; qualitative convergence.
- b.
Two-point refinement; residuals reported; limited mechanism-specific studies.
- c.
Three-point refinement with GCI; mechanism-specific local refinement; residual and monitor convergence documented.
- d.
Combined spatial and temporal refinement; conservation checks; alternative model-form spot checks; numerical uncertainty compared to comparator noise.

10. Bench comparators and metrics for each mechanism (Model S and Model T)

We designed benchtop comparators to reflect each mechanism and to produce measurements directly traceable to the QOIs. For M1, we assembled straight and tortuous segments with 120 mm test lengths, instrumented with differential pressure transducers (full scale 1000 Pa, accuracy $\pm 0.4\%$ FS, zero drift ± 0.2 Pa/h). Flow was delivered by a temperature-controlled gear pump with feedback from a coriolis flowmeter ($\pm 0.5\%$ of reading). ΔP was recorded at 50, 100, 200, and 300 mL/min for each catheter size in 4.5, 5.5, and 6.5 mm vessels, for both Newtonian and Carreau–Yasuda analog fluids. Each point was measured in triplicate. Across 42 unique operating points, the mean ΔP ranged from 38 Pa (4 Fr, 6.5 mm, 100 mL/min) to 312 Pa (6 Fr, 4.5 mm, 300 mL/min).

For M2, we developed a curvature rig with a 90-degree planar bend (radius 40 mm) and an S-bend module. The catheter was advanced over a fixed wire by a linear stage at 10 mm/s with 0.5 s dwells, as per the CFD protocol. A 6-axis force–torque sensor (resolution 0.2 mN, accuracy $\pm 1\%$ of reading) at the tip fixture recorded forces in the absence of wall contact except at designed glide pads with low friction (PTFE inserts) to isolate hydrodynamic side-loads from dry friction. Lateral force time series were obtained for each flow rate and size. Peak lateral forces ranged from 2.3 mN (4 Fr, 6.5 mm, 100 mL/min) to 14.7 mN (6 Fr, 4.5 mm, 300 mL/min).

For M3, a stenosis surrogate with 50% area reduction was placed 10 mm distal to the bend exit. Two porting configurations were tested: tip-only and two side-ports at 6 mm proximal. High-speed imaging at 2000 fps with dye injection confirmed the direction and intermittency of jets near the wall. Lateral force asymmetry was measured as the difference between peak lateral forces on opposite sides over repeated traversals with mirrored bends. Asymmetric components ranged from 1.1 to 6.2 mN depending on flow split and clearance.

For M4, the Newtonian and Carreau analogs were alternated while keeping flow rate constant. ΔP increases from non-Newtonian behavior ranged from 6.7% (300 mL/min, 6.5 mm vessel, 4 Fr) to 12.8% (100 mL/min, 4.5 mm vessel, 6 Fr). Lateral forces increased by 3–11% with the Carreau analog due to elevated near-wall viscosity in low-shear recirculation zones.

Statistical analysis compared CFD and bench results using mean absolute percentage error (MAPE), root-mean-square error (RMSE), and Bland–Altman bias with 95% limits of agreement (LoA). For ΔP in M1, Model S had MAPE 4.8% (RMSE 8.9 Pa; bias 2.1 Pa; LoA ± 12.4 Pa), and Model T had MAPE 5.4% (RMSE 10.2 Pa; bias 1.6 Pa; LoA ± 13.9 Pa) for time-averaged ΔP over pushes. For M2 peak lateral force, Model S had MAPE 16.0% (RMSE 1.9 mN; bias 0.8 mN; LoA ± 3.1 mN); Model T had MAPE 8.9% (RMSE 1.1 mN; bias 0.3 mN; LoA ± 1.9 mN). For M3 asymmetric lateral force, Model S had MAPE 18.2% (RMSE 1.2 mN); Model T had MAPE 10.7% (RMSE 0.8 mN). For M4, the ΔP increase from non-Newtonian rheology was $10.3\% \pm 2.1\%$ in bench data and $9.5\% \pm 1.7\%$ (Model S) or $10.1\% \pm 1.8\%$ (Model T) in CFD; lateral force increases were $7.8\% \pm 2.5\%$ in bench data and $6.9\% \pm 2.3\%$ (Model S) or 8.6%

$\pm 2.0\%$ (Model T).

Comparator quality and metrics — gradation.

- a. Comparisons anecdotal; no quantified uncertainty.
- b. Comparators defined; limited instrumentation; sparse coverage.
- c. Instrumented comparators with traceable uncertainties; adequate coverage across operating space; standard error metrics.
- d. Multiple comparators per mechanism; cross-checks and repeatability studies; regression and agreement analyses with uncertainty budgets.

11. Relevance of the quantities of interest to trackability and resistance endpoints

The chosen QOIs were designed to map to bench acceptance criteria and to observed differences among prototypes in functional tests. For resistance, ΔP across a 120 mm segment at 200 mL/min was used as the screening metric because it correlates with pump loading and observed flow rates during simulated infusion. Bench acceptance for this family of devices is ΔP less than 150 Pa at 200 mL/min in a 5.5 mm vessel for 5 Fr designs; thresholds scale with size. For trackability surrogates, we adopted two hydrodynamic metrics complementary to friction-based metrics used elsewhere: peak lateral side-load at bends and the integrated axial drag during the standardized push. Prototypes that showed hydrodynamic side-loads above 12 mN in the curvature rig were more likely to deviate from the wire in separate mechanical trackability tests that included wall contact and realistic friction; in a 24-run set, the correlation between peak hydrodynamic side-load and off-rail deviation rate was $r = 0.78$ ($p < 0.001$). Integrated axial drag correlated with measured push forces after removal of friction via calibration factors; the slope was 0.86 N·s per N·s predicted, with intercept compatible with zero (0.03 ± 0.04 N·s).

Because the CFD outputs are used to rank designs rather than to estimate patient-level outcomes, we defined pass/fail thresholds aligned with bench gates. A design passes the hydrodynamic side-load screen if predicted peaks are below 12 mN in the 90-degree bend at 200 mL/min and below 10 mN in the S-bend; these thresholds were set to maintain margin below bench gates that account for friction. For ΔP , pass/fail thresholds were defined per size, with a nominal 10% buffer below bench gates to account for residual model uncertainty and variability not directly modeled (e.g., minor surface roughness).

We verified that predictions at nominal conditions remain relevant across typical variances in flow rate and vessel size. Across simulated variations of $\pm 10\%$ in flow rate and ± 0.2 mm in vessel diameter, the rank order of designs by ΔP and by peak lateral force changed in fewer than 5% of pairwise comparisons, indicating robust ranking against reasonable perturbations. This



Figure 1: Schematic of benchtop comparators: (a) straight and tortuous segments for axial ΔP with pressure taps; (b) curvature rig with 90-degree bend and S-bend modules, linear stage for standardized push, and 6-axis force–torque sensor; (c) stenosis surrogate module with side-port dye visualization and high-speed camera.

property supports the use of QOIs as screening tools rather than precise deterministic predictions.

QOI-to-decision linkage strength — gradation.

- a.
QOIs do not map to decisions.
- b.
Qualitative mapping; limited empirical support.
- c.
Quantitative mapping with thresholds; correlation analyses with relevant endpoints.
- d.
Prospective thresholds; demonstrated classification performance; robustness to operating variations quantified.

12. Representativeness of validation activities to the screening context

A central tenet of this assessment is that validation activities mirror the context in which the CFD outputs will be used. The phantom geometries were designed from CTA-derived centerline statistics of peripheral arteries, spanning the interquartile range of curvature and torsion. A Kolmogorov–Smirnov test comparing the bend-radius distributions of the phantom library and the CTA dataset yielded $p = 0.62$, suggesting no significant difference across the targeted range. The selection of vessel diameters in phantoms at 4.5, 5.5, and 6.5 mm matched the quartiles of the CTA-derived diameters for candidate target vessels; thus, prototypes are screened at geometries representative of common clinical use.

Surface interactions were also mirrored for hydrodynamic measurements. While frictional effects were intentionally suppressed via PTFE inserts to isolate fluid forces for M2 and M3, separate friction characterization was performed for the same catheter coatings against phantom materials. The dynamic friction coefficient in physiologic saline at 37°C was 0.11 ± 0.03 for the coated shafts against the base phantom polymer; this value informed the adjustment between hydrodynamic-only side-load thresholds and full trackability gates that include friction. The analog fluids matched blood viscosity within 0.3 mPa·s at 100 sI for the Newtonian case and within the fitted Carreau curve across 1–1000 sI.

Instrumentation traceability and environmental controls were maintained. All measurements were performed at 37

$\pm 0.5^\circ\text{C}$, with dissolved gas levels controlled to limit bubble formation. Pressure and force sensors were calibrated against NIST-traceable standards before each run-day. Repeatability studies over five repeated runs at a midrange operating point (5 Fr, 5.5 mm vessel, 200 mL/min) yielded coefficients of variation of 1.8% for ΔP and 3.5% for peak lateral force.

Representativeness of validation activities — gradation.

- a.
Validation uses convenience models dissimilar to context.
- b.
Partial overlap with context; key aspects missing.
- c.
Validation mirrors key geometric and fluid properties; measurement conditions controlled; traceability established.
- d.
Validation spans interquartile clinical variability; surrogate materials and fluids matched quantitatively; repeatability documented and comparable to model uncertainties.

13. Test conditions — alignment with operating envelopes

The tested flow rates and vessel sizes were selected to fall within typical operating envelopes; specific procedural extremes, such as hyperemic flows or severe vasospasm, were not emulated in this phase. Deployment speeds followed the standardized push profile defined for screening. Future work may incorporate additional environmental perturbations to emulate a broader set of clinical scenarios.

14. Use error — operator-induced variability considerations

The standardized push protocol and fixture minimized variability from stage control; nevertheless, differences in catheter handling, wire support, and torque application in real use are known to affect trackability outcomes. Those aspects are outside the scope of the hydrodynamic-only comparisons reported here and are addressed in separate mechanical and integrated benchtop tests.

15. Results: credibility scorecards for Model S

The table below summarizes the credibility assessment for Model S across the four mechanisms. Scores reflect the combination of parameter evidence, numerical verification, bench comparison, decision linkage, and representativeness. The one-line basis statements repeat quantitative findings developed in the sections above.

16. Results: credibility scorecards for Model T

The table below summarizes the credibility assessment for Model T across the four mechanisms. The transient capability with moving boundaries improves fidelity for mechanisms M2 and M3 while retaining good performance for time-averaged resistance (M1).

17. Discussion

This assessment set out to determine whether two CFD models could credibly inform pre-bench triage of catheter designs by predicting flow resistance and hydrodynamic contributions to trackability. The structured application of a risk-informed framework clarified where each model is strong and where caution is warranted. The steady model (Model S) excels in predicting annular resistance under the studied conditions. Its numerical uncertainty is materially below comparator noise for ΔP , and the agreement with bench across catheter sizes and vessel diameters provides confidence for ranking designs by resistance. For lateral forces driven by bend-induced secondary flows, Model S captures trends but underpredicts peaks due to the omission of unsteady interactions that emerge during motion. The transient model (Model T) reduces this gap by incorporating time-dependent tip-flow interactions, moving meshes, and the standard push protocol, resulting in improved prediction of peak side-loads and of asymmetric forces in jetting scenarios.

A noteworthy aspect is the alignment of the QOIs with internal design gates. Decision thresholds were pre-specified for ΔP and hydrodynamic side-loads, and the comparators were configured to measure the same metrics. This affords direct use of CFD outputs in binning designs without ad hoc translation. Designs identified as acceptable *in silico* were, in the majority of cases, the same that passed bench gates, with classification agreement of 93% across 28 prototypes for lateral-force thresholds. Such agreement does not imply that the CFD replaces bench tests, nor that it can be used outside the defined operating space; it does support its role as an efficient, lower-cost filter.

The parameter evidence and sensitivity analyses also guide how to maintain credibility as designs evolve. For annular resistance, small changes in vessel diameter dominate ΔP ; therefore, maintaining tight control of phantom dimensions and accurately characterizing target vessel sizes in new screening programs is more impactful than refining viscosity measurements within very narrow bands. For bend-induced lateral forces and jetting asymmetries, geometric details of bends, tip clearance to walls, and port flow splits matter most. This suggests that when

novel porting geometries are introduced, renewing the port-flow characterization and repeating M3 comparisons would be a priority.

We emphasize the separation of numerical, parameter, and model-form uncertainties. For both models, the mesh and time-step studies drove numerical uncertainty below the levels that influence decisions. Therefore, when differences between model and bench appear in mechanisms M2 and M3, they are better attributed to the simplifications inherent to steady flow (Model S) or to residual uncertainties in port flow splits and clearances, rather than to discretization. This separation prevents misguided efforts to chase numerical artifacts when the primary gaps are physical or parameter-related.

The validation activities were designed to mirror the screening context rather than to emulate every clinical nuance. For example, wall friction was intentionally suppressed in hydrodynamic side-load measurements to reduce confounding; separate mechanical tests integrate friction and contact. Likewise, the stenosis surrogate was limited to a single severity (50% area reduction) for this phase; future work can extend to a range of severities to bound how often port-jet asymmetries reverse or amplify. The analog fluids matched target rheology profiles closely, but we did not include hyperemic extremes or time-varying flows typical of pulsatility; because the CFD is used for screening under nominal flows, this limitation is acceptable.

The observed dispersion in M2 for the steady model highlights a familiar lesson: local unsteadiness can matter even when Reynolds numbers are not classically turbulent. The transient model's explicit treatment of advancement, with conservative time-step and mesh strategies, reduced MAPE to below 10% for peak side-load. The remaining differences likely arise from small, difficult-to-measure clearances and from three-dimensional features such as secondary torsion of the path and guidewire deflection, which were minimized but not eliminated in our fixtures. Notably, the sign and magnitude of asymmetric lateral forces in M3 were correctly predicted by the transient model across both tip-only and side-port configurations; this is important because misprediction of the sign would undermine any threshold-based screening.

For hemorheology, both models captured the modest but non-negligible increases in ΔP and lateral forces when moving from Newtonian to Carreau-Yasuda analogs. The magnitude of change was similar across models and matched bench within one percentage point for ΔP . This suggests that under the studied flows and geometries, first-order viscosity effects dominate and detailed shear-thinning profiles make a predictable, monotonic contribution. In contrast, in side-loads near recirculation, the effect is smaller and may be masked by geometric variability unless carefully isolated.

A pragmatic outcome of this assessment is prioritization of future validation. If transient simulations are to replace steady ones for all screening, additional model management will be required because they are costlier. A hybrid strategy may be optimal: Model S for rapid ΔP screening and for coarse side-load triage, with Model T applied to borderline or high-priority designs where hydrodynamic side-loads approach thresholds or

where jetting asymmetry is a critical feature.

18. Implications for design decisions within the stated context

Across mechanisms and models, the chosen parameter sets were confined to ranges supported by imaging, metrology, and rheometry, and their combined uncertainty propagated to the reported outputs yielded at most an 8% spread in the screening metrics at nominal operating points, which is inside the acceptance bands used to separate design bins.

Discretization and iteration settings were strong enough that the remaining numerical fringes sat below the measurement noise of the bench comparators, not exceeding 1.0% for pressure loss and 3.5% for peak lateral-force predictions, so we did not ascribe data–model differences to mesh or time-step insufficiency.

When paired against the laboratory reference, median absolute percentage differences were 5.2% for pressure loss and 9.1% for peak lateral force across models and mechanisms, with no significant slope bias versus flow rate, catheter size, or vessel diameter in Passing–Bablok regressions.

The selected reporting metrics map directly to the go/no-go decision—designs that met the side-load threshold of 12 mN and the 150 Pa at 200 mL/min resistance threshold on bench were consistently the same designs that met the CFD criteria in 93% of 28 prototypes, enabling confident binning without ad hoc translation.

The phantoms, analog fluids, and deployment paths were tuned to the screening landscape: the shapes spanned the interquartile tortuosity of 176 CTA-derived vessels, and surface interactions matched coated shafts within 12% friction, which makes the evidence gathered appropriate to the intended use.

19. Limitations confined to the assessed factors

The scope of this credibility assessment is restricted to hydrodynamic mechanisms and to quantities of interest that feed the pre-bench design screen. We did not model or validate frictional contact, wire–wall interactions, or torque-transmission effects, though we controlled or measured them in bench setups to isolate fluid forces. The validation activities were limited to nominal, steady or quasi-steady flow rates; pulsatility and pressure waveforms were not included. While transient advancement was modeled and validated, deformation of the catheter shaft was not coupled to the flow in this phase; the moving boundary was kinematically prescribed, and structural flexibility was neglected. This is acceptable for small deflections under the studied conditions but may not hold for very soft distal sections or highly tortuous paths.

The stenosis surrogate was limited to a 50% area reduction. Different severities, eccentricities, or lesion lengths could change local jet behavior and, therefore, the magnitude and direction of asymmetric lateral loads. Future work should extend to a small set of additional severities to check for sign changes and nonlinearities that might influence screening decisions for highly ported tips.

Rheology was modeled for a 45% hematocrit analog. Clinical hematocrit varies; although sensitivity analyses suggest modest changes in ΔP and lateral forces across plausible ranges, direct validation at additional hematocrits would strengthen generality if the screening program is to be applied to populations with marked deviations.

Finally, although the phantom library was designed to span the interquartile range of tortuosity and diameter, the extremes of the distribution were not covered. For rare anatomies at the tails, screening decisions should be interpreted with caution, and bench tests with tailored phantoms are advised before drawing conclusions.

20. Conclusions

Using ASME V&V 40's risk-informed approach, we assessed the credibility of two CFD models intended to screen over-the-wire peripheral catheters for hydrodynamic contributors to trackability and for hydraulic resistance. Under a medium-risk context of use, we found that the steady CFD model is well suited to ranking designs by annular pressure-drop and to providing directional insight into side-load trends at bends. The transient model improved fidelity for peak lateral side-loads and asymmetric forces due to jetting while maintaining accurate time-averaged resistance predictions. Both models captured rheology-driven trends. The five-factor, mechanism-level scorecards provide concise, quantitative support for these conclusions and make explicit where additional validation would most improve confidence. The practical implication is a hybrid screening strategy: rapid steady simulations for resistance and broad triage, and targeted transient simulations for designs near side-load thresholds or with complex porting.

As simulation capabilities and validation resources evolve, this assessment can be updated by adding new mechanisms, expanding the operating space, and refining evidence depth commensurate with any change in decision influence. In its current form, the evidence is sufficient to support the intended use: informing design triage while maintaining reliance on benchtop testing for verification and on clinical evidence for performance in patients.

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Table 1: Credibility Index (CI) for Model S across mechanisms M1–M4. Scale: 0–5.

Credibility factor	Level	Basis
Model inputs - Model S - M1 Axial annular pressure-drop	4	Vessel and catheter diameters measured to $\pm 0.05/\pm 0.02$ mm; Sobol' S1 for vessel diameter 0.62; ΔP sensitivity $\pm 7.1\%$ per ± 0.05 mm; viscosity bounded within ± 0.2 mPa·s.
Numerical solver error - Model S - M1 Axial annular pressure-drop	5	Three-mesh refinement (1.2, 3.6, 9.6M cells) gave GCI $\leq 0.6\%$ for ΔP ; residuals $< 1e-6$; domain-length sensitivity $< 0.3\%$.
Output comparison - Model S - M1 Axial annular pressure-drop	4	Against 42 points, MAPE 4.8%, RMSE 8.9 Pa, bias 2.1 Pa, LoA ± 12.4 Pa over 38–312 Pa range.
Relevance of the quantities of interest - Model S - M1 Axial annular pressure-drop	5	ΔP at 200 mL/min aligns with screening gate (150 Pa for 5 Fr); rank order robust to $\pm 10\%$ flow and ± 0.2 mm diameter with $< 5\%$ pairwise reversals.
Relevance of validation activities to the COU - Model S - M1 Axial annular pressure-drop	4	Phantom diameters 4.5/5.5/6.5 mm match CTA quartiles; fluid viscosity within 0.3 mPa·s at 100 sI; repeatability CV 1.8%.
Model inputs - Model S - M2 Dean-type secondary flows at bends	3	Bend radii 35–45 mm trace to CTA IQR; eccentricity 0.2–0.5 mm explored; peak side-load sensitivity 11%/+9% for ± 5 mm radius, +7–10% for 0.5 mm offset.
Numerical solver error - Model S - M2 Dean-type secondary flows at bends	4	Local refinement to 80 μ m; mesh counts 2.4/6.8/12.5M; lateral force GCI 2.7–3.8%; residuals $< 1e-6$.
Output comparison - Model S - M2 Dean-type secondary flows at bends	3	Peak lateral force MAPE 16.0% (RMSE 1.9 mN; bias 0.8 mN; LoA ± 3.1 mN) across 2.3–14.7 mN.
Relevance of the quantities of interest - Model S - M2 Dean-type secondary flows at bends	4	Hydrodynamic side-load threshold 12 mN differentiates designs; correlation to off-rail deviation $r=0.78$ in 24-run set.
Relevance of validation activities to the COU - Model S - M2 Dean-type secondary flows at bends	3	Curvature modules mirror CTA IQR (KS $p=0.62$); PTFE pads isolate fluid forces; force sensor accuracy $\pm 1\%$.
Model inputs - Model S - M3 Tip and side-port jetting with recirculation	3	Port geometry measured (0.4 mm) and tip/port split varied by ± 8 pp; asymmetric force sensitivity index 0.41 (split) and 0.33 (gap).
Numerical solver error - Model S - M3 Tip and side-port jetting with recirculation	4	Tip/port mesh down to 80 μ m; outflow extension 8D; lateral force GCI 3.1%; LES spot check changed peaks $< 2\%$.
Output comparison - Model S - M3 Tip and side-port jetting	2	Asymmetric lateral force MAPE 18.2% (RMSE 1.2 mN); systematic

Table 2: Credibility Index (CI) for Model T across mechanisms M1–M4. Scale: 0–5.

Credibility factor	Level	Basis
Model inputs - Model T - M1 Axial annular pressure-drop	4	Geometries and fluids as in Model S; advancement protocol 40 mm at 10 mm/s; ΔP sensitivity to diameter and viscosity as in M1; time-averaging window defined by push.
Numerical solver error - Model T - M1 Axial annular pressure-drop	4	Mesh 1.8/5.5/14M; dt 1.0/0.5/0.25 ms; time-averaged ΔP uncertainty $\leq 1.2\%$; residuals $< 1e-5$; mass conservation within 0.05%.
Output comparison - Model T - M1 Axial annular pressure-drop	4	MAPE 5.4% (RMSE 10.2 Pa; bias 1.6 Pa; LoA ± 13.9 Pa) for averaged ΔP over push in 42 points.
Relevance of the quantities of interest - Model T - M1 Axial annular pressure-drop	5	Averaged ΔP aligns with pump loading during motion; thresholds mirror bench gates with 10% buffer; ranking robust to flow/diameter perturbations.
Relevance of validation activities to the COU - Model T - M1 Axial annular pressure-drop	4	Same phantoms and fluids; temperature-controlled; repeatability CV 1.8% for ΔP ; same flow-rate controls.
Model inputs - Model T - M2 Dean-type secondary flows at bends	4	Bend radii 35–45 mm; eccentricity 0.2–0.5 mm; sensitivity shows peak force 9–+11% for ± 5 mm radius; protocol prescribes dwell and speed to match bench.
Numerical solver error - Model T - M2 Dean-type secondary flows at bends	4	Peak lateral force changed 3.6% (dt 1.0 to 0.5 ms) and 1.2% (0.5 to 0.25 ms); mesh refinement to 14M; residuals $< 1e-5$.
Output comparison - Model T - M2 Dean-type secondary flows at bends	4	Peak lateral force MAPE 8.9% (RMSE 1.1 mN; bias 0.3 mN; LoA ± 1.9 mN) across 2.3–14.7 mN.
Relevance of the quantities of interest - Model T - M2 Dean-type secondary flows at bends	5	Peak hydrodynamic side-load maps to 12 mN threshold; classification agreement with bench 93% across 28 prototypes.
Relevance of validation activities to the COU - Model T - M2 Dean-type secondary flows at bends	4	Curvature modules within CTA IQR; PTFE isolation; force sensor accuracy $\pm 1\%$; stage motion emulates protocol used in CFD.
Model inputs - Model T - M3 Tip and side-port jetting with recirculation	3	Port sizes 0.4 mm; nominal 70/30 split; sensitivity to split ± 8 pp; stenosis 50% AR; gap-dependent asymmetry captured.
Numerical solver error - Model T - M3 Tip and side-port jetting with recirculation	4	Local remeshing maintains 70–120 μ m near tip/ports; peak force change 1.4% with dt halved; residuals $< 1e-5$.
Output comparison - Model T - M3 Tip and side-port jetting	4	Asymmetric lateral force MAPE 10.7% (RMSE 0.8 mN); correct sign and